

Network Hardening — Guided Notes

Unit 4.1.3 | CompTIA Network+ N10-009 Obj. 4.2 | N10-008 4.2

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

Network hardening means reducing the _____ surface by disabling unused services, ports, and accounts. List three specific actions that reduce attack surface on a managed switch.

Question 2

Port security limits the number of _____ addresses allowed on a switch port. When a violation occurs, the three response modes are _____ (drops frames, no alert), _____ (drops frames, logs alert), and _____ (disables the port).

Question 3

802.1X uses three roles: the _____ (client device), the _____ (switch or AP that relays authentication), and the _____ (RADIUS server that validates credentials). What does this prevent that MAC filtering cannot?

Question 4

BPDU Guard protects against rogue switches being added to the network by disabling any access port that receives a _____. This prevents a connected device from participating in _____ elections.

Question 5

DHCP snooping builds a binding table of _____ IP-to-MAC-to-port mappings and drops DHCP server messages from _____ ports. Which type of attack does this directly prevent?

Question 6

Dynamic ARP Inspection (DAI) uses the DHCP snooping binding table to validate ARP packets. It drops ARP replies where the _____ does not match the binding table entry. This defeats _____.

Question 7

A network baseline documents normal behavior (traffic volumes, protocols, connection patterns) so that _____ can be identified. Name two specific metrics that a baseline should capture.

Question 8 — Real World Application

REAL WORLD: A student plugs a cheap consumer router into a classroom Ethernet jack. The router's DHCP server starts handing out 192.168.1.x addresses, overriding the school's DHCP. Name the attack type, the switch feature that would have prevented it, and the command category used to implement it.

Download the answer key from the class server:

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wget http://[SERVER]/resources/network-engineer/4-1-3-network-hardening/4-1-3-network-hardening-answer-key.pdf
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